Race, explain and transfer

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Evaluate a model against measured performance and defend design decisions.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 8, Day 3.*
Ideal speed becomes 6 times as large. Actual rpm, load or slipping may change.

02. Worked example: explain the model
Electrical input to motor/mechanical motion plus thermal/sound transfers; actual measurements distinguished from illustrative values.

03. Read and interpret the model
Three supported explanations using relevant mechanics/circuit ideas.

04. Apply the physics
0.50 m/s; 16.7% below the prediction.

05. Explain your reasoning
Actual evidence and bounded explanations; a lower race placing alone does not diagnose a cause.

06. Review: connect an earlier idea *Review: Unit 7, Day 15.*
A generator. Some energy can become heating, friction or sound.

07. Exit check
Student names a concrete next step tied to evidence or a skill; self-rating is not a teacher score.

08. STRETCH: test the model
Example lower gearing may improve starting torque but reduce unloaded speed; test start reliability and timed speed under controlled conditions.

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01. Warm-up: bring back an earlier idea *Review: Unit 8, Day 3.*
Torque ratio 8; speed ratio 1/8. Assumes ideal gears with negligible losses.

02. Worked example: explain the model
Electrical input to motor/mechanical motion plus thermal/sound transfers; actual measurements distinguished from illustrative values.

03. Read and interpret the model
Three supported explanations using relevant mechanics/circuit ideas.

04. Apply the physics
0.75 m/s; 16.7% below prediction; load, slip, friction or model assumptions, with evidence needed to attribute cause.

05. Explain your reasoning
Actual evidence and bounded explanations; a lower race placing alone does not diagnose a cause.

06. Review: connect an earlier idea *Review: Unit 7, Day 15.*
Combined fraction=0.4608 or 46.08%, multiplying the two efficiencies.

07. Exit check
Ideal speed doubles; increased required torque, loading, slip and clearance may change actual rpm/performance.

08. STRETCH: test the model
Input 24 J; K 1 J; ratio≈4.17%. Also account for rotational energy, work against resistance, stored energy and time history.